Orthogonally quantifying chemical contamination; Validating least-contaminated (reference) sites; Identifying and extending natural community clusters from reference sites to all sites.

Summaries of Chapter 2 & 3

Feng Gu

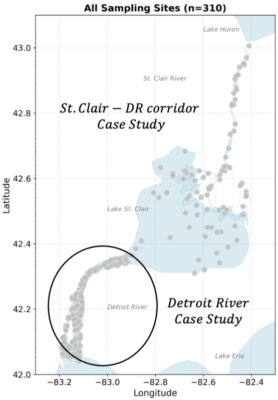
Progress Update

Overview of the Whole Thesis

Core target: study taxa community distortion extent in terms of contamination levels.

- **Chapter 1** — Quantify **contamination levels** of sites; Validate least-contaminated sites where contamination has minimal impact on community composition.
- **Chapter 2** — Identify natural **taxa clusters** in least-contaminated sites; Assign the natural clusters to all sites by environmental descriptors, equivalently partitioning natural variation in contaminated sites.
- **Chapter 3** — In each community type, measure the **ecological dissimilarity** among sites; scale the dissimilarity among sites into an coordinate system where euclidean distance mostly preserves that dissimilarity.
- **Chapter 4** — In each community type, apply regression analysis (piecewise quantile regression) between the scaled community dissimilarity and contamination levels; detect **ecological threshold(s)** and track back to finer-level taxa and latent stressors.

Study Area and Data Source



Data Source	Symbol	Shape	Unit
Environmental	$\boldsymbol{E}$	(310×6)	array (6×1)
Chemical	$\boldsymbol{M}$	(310×16)	array (16×1)
Taxa	$\boldsymbol{T}$	(310×16)	$octave\ transformed$

- Columns/Features in $\boldsymbol{E}$: Temperature, Water Depth
- Features in $\boldsymbol{M}$: Al, As, OCS ...
- Features in $\boldsymbol{T}$: Dressina, Nematoda ...

- There are two case studies in this thesis: (1) St. Clair - Detroit River corridor case study; (2) Detroit River case study.
- The same methodology is applied to both case studies, DR case study uses the subset of the data in the corridor case and has one additional environmental feature — estimated *water velocity*.

Chapter 1 — Validating the Selection of Least-Contaminated Reference Sites

Chapter 1 at a Glance: Background, Method, Result

Background

Quantifying contamination level is essential for regressing their effects on taxa and selecting least-contaminated sites to find *natural* community types as benchmarks.

Method

Reduce 16 correlated chemicals to orthogonal latent stressors (PCA), sum them into an integrated score (SumRel), then *test* the least-contaminated set cut-off number with two complementary resampling tests.

Result

The lowest-20% set is significantly cleaner than random sets and carries no internal contamination constraint; a size scan shows the corridor tolerates 20% while the Detroit River does not.

Why “Least-Polluted” Is the Only Defensible Baseline

Natural vs. least-polluted

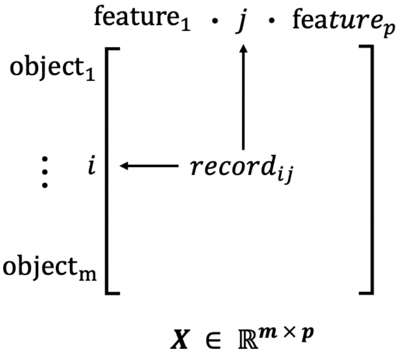
A **natural site** is a theoretical reference state shaped without human stressors — not directly observable here. A **least-polluted site** is operational: defined relative to the observed distribution of integrated chemical stress, taking its low tail as the baseline.

- Within least-polluted sites, community structure is governed by habitat, *not* chemical stress — they retain naturally organised community information.
- They are atoms to constitute the comparable, observable ecological benchmarks for the rest of the thesis.

From 16 Correlated Chemicals to Orthogonal Latent Stressors

- The 16 chemicals are ecologically distinct but **statistically redundant**; correlated variables would carry overlapping information and double-count.
- PCA (R-mode ordination, Euclidean distance) extracts **latent stressors**: mutually orthogonal axes preserving the original stress information.
- Orthogonality guarantees each stressor's signal is uncorrelated with the others, so contamination variation is attributed to the right domain.

Matrix Data Structure: record $\mathbf{X}_{ij}$ stores object i's value of feature j via pointing paths



How we view a matrix structure in this thesis

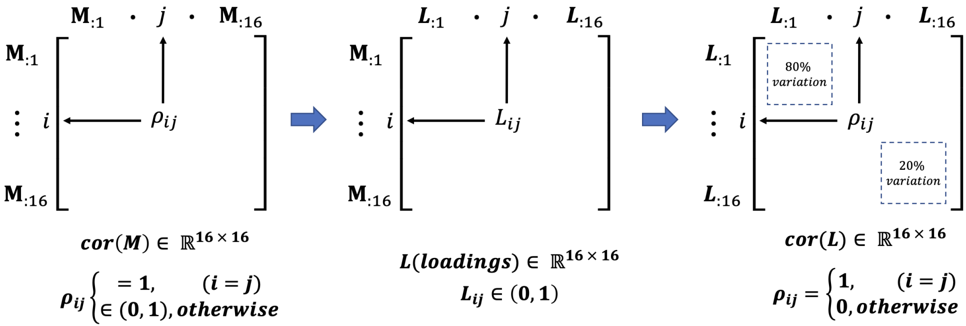
$\mathbf{X} \in \mathbb{R}^{m \times p}$ stores m objects' records of p features:

- Row dimension has m ticks: object i is the i-th tick.
- Column dimension has p ticks: feature j is the j-th tick.
- Entry X_{ij}: a record pointing to object i and feature j.
- $X_{:j}$: the vector of m records pointing to all m objects for feature j.

This *record-pointing* view is used throughout the thesis and later extends to higher-dimensional tensor structures[1].

[1] A matrix is a 2-dimensional tensor: two dimensions store the ticks a record points to, so $X_{ij} = \mathbf{X}[i, j]$.

Source of overlapping information in chemical matrix **M** and the solution



- In the **M** matrix, features $\mathbf{M}_{:i}$ and $\mathbf{M}_{:j}$ are correlated; so the recorded information is double-counted, shown by: $Var(\sum_{i=1}^{16} \mathbf{M}_{:i}) = \sum_{i=1}^{16} Var(\mathbf{M}_{:i}) + 2\sum_{i<j} Cov(\mathbf{M}_{:i}, \mathbf{M}_{:j})$ [2] .
- In the resulted loading matrix **L** and PC score matrix $\mathbf{S} = \mathbf{ML}$, $\mathbf{L}_{:i}$ and $\mathbf{L}_{:j}$ are orthogonal and likewise to $\mathbf{S}_{:i}$ and $\mathbf{S}_{:j}$ [3] so $Var(\sum_{i=1}^{p} \mathbf{S}_{:i}) = \sum_{i=1}^{p} Var(\mathbf{S}_{:i})$.

[2] This overlapping-information structure is specifically studied in the field of time series analysis.

[3] $\mathbf{L}_{:i}$ and $\mathbf{L}_{:j}$ are orthogonal eigenvectors of the correlation matrix $\mathbf{cor}(\mathbf{M}) = \frac{1}{n-1}\mathbf{M}^T\mathbf{M}$ (with **M** standardised), so $\mathbf{cor}(\mathbf{M})\mathbf{L}_{:j} = \lambda_j \mathbf{L}_{:j}$. Hence $\mathbf{S}_{:i}^T\mathbf{S}_{:j} = \mathbf{L}_{:i}^T(\mathbf{M}^T\mathbf{M})\mathbf{L}_{:j} = (n-1)\lambda_j\,\mathbf{L}_{:i}^T\mathbf{L}_{:j} = 0$.

Computation steps from chemical matrix **M** to PC Loading matrix-**L** and PC scores matrix-**S**

Algorithm 1: Construction of the PC-based chemical stressor matrix

Data: Site-by-chemical matrix $\mathbf{M}_{310\times 16}$.

Result: PC-based chemical stressor matrix $\mathbf{S}_{310\times p}$.

Log-transform chemical matrix and compute its correlation matrix:

$$\mathbf{M}_{310\times 16} \rightarrow \mathbf{M}'_{310\times 16} \xrightarrow{\text{cor}} \mathbf{cor}(\mathbf{M}')_{16\times 16}$$

Apply PCA on correlation matrix, quartimax rotation (flip if needed), and loading selection:

$$\mathbf{cor}(\mathbf{M}')_{16\times 16} \xrightarrow{pca} \mathbf{Loadings}_{16\times p'} \xrightarrow{\text{rotation}} \mathbf{Loadings}^{\star}_{16\times p'} \xrightarrow{\text{check \& selection}} \mathbf{L}^{\star}_{16\times p}$$

Standardize columns in $\mathbf{M}'$, project the 16 processed chemical variables to the loading axes:

$$\mathbf{M}'_{310\times 16} \xrightarrow{standardize} z(\mathbf{M}')_{310\times 16};\ \mathbf{S}_{310\times p} = z(\mathbf{M}')_{310\times 16}\mathbf{L}^{\star}_{16\times p}$$

return $\mathbf{S}_{310\times p}$

- Log-transformation compresses the extreme values so that they do not improportionally distort the correlation coefficients $\mathbf{cor}(\mathbf{M})_{ij}, (i \neq j)$ between chemical variables.
- Loading rotation concentrates each PC's loadings on a few chemical variables, so PCs can be interpreted more directly in terms of those dominant chemicals.

Drafts

Resulting PC Loading matrix $\mathbf{L}^*$ (Corridor case)

Chemical	**PC_1**	**PC_2**	**PC_3**	**PC_4**	**PC_5**	**PC_6**
Cr	**0.9513**	0.1267	0.1178	0.0032	−0.0226	0.0222
Cu	**0.8999**	0.0993	0.0920	−0.0732	0.0184	0.0117
Ni	**0.8819**	−0.0435	−0.0654	0.0386	−0.2756	0.0631
Zn	**0.8707**	−0.1695	−0.0187	−0.0708	0.1240	−0.0418
Fe	**0.7923**	0.1698	0.0292	0.0245	0.0519	0.2347
SumPCBs	**0.7756**	−0.4605	0.0212	0.1543	0.0438	−0.0751
Co	**0.7635**	0.4566	0.0062	0.0725	0.3474	0.0609
Cd	**0.7358**	−0.2001	0.4249	0.0515	0.1110	0.0586
p,p'-DDE	**0.5937**	−0.3834	−0.0932	0.3676	0.2515	−0.1579
Mn	**0.5897**	0.4149	0.3051	0.1630	−0.2672	0.3835
Al	0.5676	**0.6718**	0.0079	0.0345	0.3298	0.1848
As	0.3392	−0.0176	**0.8093**	−0.0294	0.2989	0.0364
Pb	0.3933	0.0837	**0.7681**	0.0369	−0.3819	−0.0015
OCS	0.1905	0.0145	0.0149	**0.9444**	0.0122	0.1100
Hg	0.4137	0.1400	0.0236	0.0359	**0.8039**	0.0486
Ca	0.2978	0.0942	0.0199	0.0965	0.0466	**0.9299**
Explained Var.	7.1925	1.3524	1.5568	1.1119	1.3571	1.1609
Proportion of Var.	0.4495	0.0845	0.0973	0.0695	0.0848	0.0726
Cumulative Var.	0.4495	0.5341	0.6313	0.7008	0.7857	0.8582

Stressor Matrix **S** and Collapsing Stressors into One Integrated Score

Of the first six rotated PCs, four, loading high on **toxic** chemicals, are retained in $\mathbf{L}^*$. PC_2 and PC_6 are dropped: they load positively on the geogenic elements Al, Ca, and Fe [1], which reflect **natural** lithogenic background rather than contamination [2], and negatively on some toxic chemicals (p,p'-DDE, SumPCBs). Projecting the processed chemical matrix $\mathbf{M}'$ gives four latent stressors per site:

$$\mathbf{S}_{310\times 4} = \mathbf{M}'_{310\times 16}\,\mathbf{L}^*_{16\times 4}$$

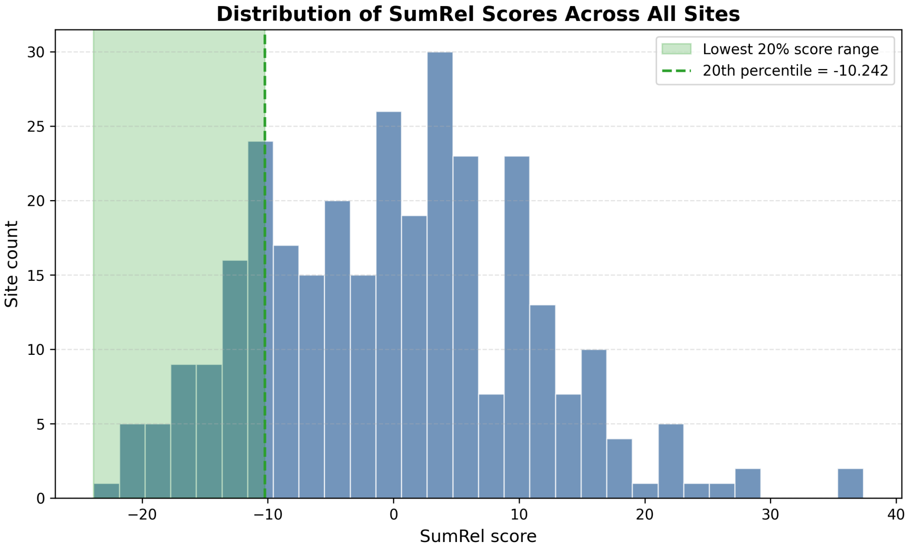
which are summed into one integrated score:

$$\mathrm{SumRel}_i = \sum_{j=1}^{4} s_{ij}, \qquad s_{ij} \in \mathbf{S}_{310\times 4}$$

Integrated contamination level and the additive-impact perspective

Summing the latent stressors into SumRel collapses them into a single measure of *integrated* contamination level. Studying the community response against this sum *implicitly assumes* an *additive impact* [3]. Alternative impact views [4] exist, but this study adopts the additive perspective.

Resulting SumRel Histogram and a candidate least-contaminated set



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Bridge the Gap between Least-Contaminated and Natural Benchmark

Background and motivation

A 20% lowest cut-off is conventionally used to select least-contaminated sites, they are assumed minimally impacted and set as natural benchmarks. Variously contaminated areas have their lowest 20% sites impacted differently. It needs to test whether a cut-off value choses sites of defensible natural benchmarks and how to adjust it if it did not.

Objectives

- Given a cut-off m_{ref}, answer whether the lowest-scored m_{ref} sites have their taxa community significantly less impacted than a random m_{ref}-unit subset.
- Given a m_{ref}-unit set, answer whether its community is significantly affected by stressor levels. And how likely a random m_{ref}-unit set is affected by stressor levels.
- Answer how the cut-off m_{ref} and the population-level contamination effect simultaneously deciding if a least-contaminated set is defensible as a natural benchmark.

Data, Methods and Theoretical Support

- Taxa matrix $\mathbf{T}$, Stressor matrix $\mathbf{S}$
- Multivariate Regression; adjusted R^2; Resampling tests; Permutation tests.
- Central Limit Theorem; Monte Carlo Simulation; DKW inequality.

Between-group hypothesis test: Are the least-contaminated m_{ref} communities less impacted than random m_{ref} communities?

Hypotheses for the between-group impaction test

H_i	**Statement**
H_0	The cut-off m_{ref} gives a least-contaminated set η_{ref} that does not explain less community variation than a random m_{ref}-unit subset η_i.
H_A	The cut-off m_{ref} gives a least-contaminated set η_{ref} that explains less community variation than a random m_{ref}-unit subset η_i.

Steps of the test

- Set a cut-off $m_{\textbf{ref}}$ and fetch least-contaminated set η_{ref}. Fit multivariate regression between their taxa composition and latent stressors, $\hat{\mathbf{T}}_{\eta_{\text{ref}}} = \mathbf{S}_{\eta_{\text{ref}}}\beta$. The stressor-explained community variation fraction is $\bar{R}^2_{\textbf{ref}} = \frac{SS_{\textbf{fit}}}{SS_{\textbf{tot}}}$ after dimensionality adjustment.
- Randomly draw m_{ref} units from the $N = 310$ sites, repeat B times to form a set of random subsets $\{\eta_i\}_{i=1}^{B}$. Fit the same regression $\hat{\mathbf{T}}_{\eta_i} = \mathbf{S}_{\eta_i}\beta$ for all i and collect the explained variation fractions into $\{\bar{R}^2_i\}_{i=1}^{B}$.
- Compare $\bar{R}^2_{\text{ref}}$ with the **null distribution** $\{\bar{R}^2_i\}_{i=1}^{B}$, if $\bar{R}^2_{\text{ref}}$ lies significantly below $\{\bar{R}^2_i\}$, the reference set η_{ref} is significantly less impacted than a random m_{ref}-size subset.

Within-group (permutation) hypothesis test: Is a fixed set of m_{ref} communities internally affected by contamination?

Hypotheses for the within-group impaction test

H_i	**Statement**
H_0	The m_{ref} sites have their community variation not internally affected by stressor levels.
H_A	The m_{ref} sites have their community variation internally affected by stressor levels.

Steps of the test

- To one m_{ref}-unit set η_i, fix $\hat{\mathbf{T}}_{\eta_i}$ rows , permute $\mathbf{S}_{\eta_i}$ rows with B' times to generate permuted pairs $\{\hat{\mathbf{T}}_{\eta_i}, \mathbf{S}^{\pi}_{\eta_i}\}_{\pi=1}^{B'}$. Fit multivariate regression $\hat{\mathbf{T}}_{\eta_i} = \mathbf{S}^{\pi}_{\eta_i}\beta$ for all permuted B' pairs and collect the explained variation fractions into $\{\bar{R}^2_{\pi}\}_{\pi=1}^{B'}$.
- Compare η_i's observed $\bar{R}^2_{\eta_i}$ with its **null distribution** $\{\bar{R}^2_{\pi}\}_{\pi=1}^{B'}$, it fails to reject H_0 if $\bar{R}^2_{\eta_i}$ is not significantly higher than $\bar{R}^2_{\pi}$ with a right-tailed p-value $< \alpha$.
- There are B random m_{ref}-unit sets $\{\eta_i\}_{i=1}^{B}$. Each set η_i is tested with B' permutations, and produces a $\{\bar{R}^2_{\pi,\eta_i}\}_{\pi=1}^{B'}$ with its p_{η_i}-value. The $\{p_{\eta_i}\}_{i=1}^{B}$ shows how likely a random m_{ref}-unit set fails to reject H_0 or not.

$\hat{\bar{R}}^2_m$ & $\hat{\bar{R}}^2_\pi$: Asymptotic Normality by CLT and DKW Estimation Accuracy

Between-group test. $\bar{R}^2$ is a smooth function of **linear** cross-product quantities computed over the N-unit population[1]. By the finite-population CLT [5] and the Delta method [6], the estimate $\hat{\bar{R}}^2_m$ from a random m-unit subset is **asymptotically normal** around the population-level value $\bar{R}^2$:

$$\sqrt{m}\left(\hat{\bar{R}}^2_m - \bar{R}^2\right) \xrightarrow{d} \mathcal{N}(0, \tau^2)$$

For a fixed m, drawing B random m-unit subsets out of the $\binom{N}{m}$ possible sets produces an empirical CDF $\hat{F}_{m,B}(t)$ **approximating** the true subset-level CDF $F_m(t)$. This Monte Carlo approximation [7] has error bounded by the DKW inequality [8, 9]:

$$P\left(\sup_t \left|\hat{F}_{m,B}(t) - F_m(t)\right| > \varepsilon\right) \leq 2\, e^{-2B\varepsilon^2}$$

Within-group test. Fixing one m-size set, there are $m!$ row-permutations. By Hoeffding's combinatorial CLT [10], a single permutation estimate R_π, standardized by the exact closed-form permutation moments μ_S and σ_S of this one set, is **asymptotically standard normal**:

$$\frac{R_\pi - \mu_S}{\sigma_S} \xrightarrow{d} \mathcal{N}(0, 1)$$

where μ_S and σ_S are the mean and standard deviation of R_π taken over all $m!$ permutations of this set. Sampling B' permutations similarly gives an empirical permutation-null $\hat{G}_{m,B'}(t)$ approximating the exact permutation-null $G_S(t)$.

[1] $N = 310$ (corridor) or 213 (DR) is fixed in this study; the statement is used as a finite-sample approximation, justified by m and $N - m$ both being reasonably large, rather than a literal double limit.

Testing Results: Determined by Both Cut-off m_{ref} and Population-Level $\bar{R}^2$

Between-group test. As the cut-off m_{ref} **increases**, the null distribution $\hat{\bar{R}}^2_m$ concentrates more tightly around $\bar{R}^2$ (variance $\sim \tau^2/m$ from the Delta method below), so a given gap between $\bar{R}^2_{\text{ref}}$ and the null mean becomes more standard deviations wide, making H_0 **easier to reject**:

$$\sqrt{m}\left(\hat{\bar{R}}^2_m - \bar{R}^2\right) \xrightarrow{d} \mathcal{N}(0, \tau^2)$$

Conversely, as m_{ref} **decreases**, the null distribution spreads out, and a larger gap is needed to reach the same significance level, making H_0 **harder to reject**. (This holds for a fixed observed gap; in practice $\bar{R}^2_{\text{ref}}$ itself also shifts with m_{ref}, since the reference set's composition changes.)

Within-group test. For one fixed set (size m_{ref}, observed value $\bar{R}^2_{\text{ref}}$), the permutation test gives, for π a random permutation among the $m_{\text{ref}}!$ total:

$$\frac{R_\pi - \mu_S}{\sigma_S} \xrightarrow{d} \mathcal{N}(0, 1)$$

Since the stressor variables are centered, $\mu_S \approx 0$ regardless of m_{ref} and only σ_S depends on m_{ref}. As m_{ref} **increases**, σ_S decreases and a tighter permutation null makes it **harder to not reject H_0**. As m_{ref} **decreases**, σ increases, and a wider permutation null makes it **easier to not reject H_0**.

Takeaway

Whether the least-contaminated m_{ref} set passes both tests depends jointly on the cut-off m_{ref} and the population-level $\bar{R}^2$: a less contaminated study area and/or a smaller m_{ref} make both tests easier to pass; a more heavily contaminated area and/or a larger m_{ref} make them harder.

Putting It Together — Scanning the Threshold

Both tests are run at every candidate size, sweeping the least-contaminated set from 15 to 310 sites. The size at which p_{ref} falls below α reveals where contamination begins to constrain the community.

Algorithm 2: Threshold scan across least-contaminated subset sizes

Data: Stressor $\mathbf{X}$, community $\mathbf{Y}$, SumRel-ordered sites, step $\delta = 5$.

Result: Curves $\bar{R}^2_{\text{ref}}(m_{\text{ref}})$ and $p_{\text{ref}}(m_{\text{ref}})$ with bands.

for $k \leftarrow 0$ **to** 59 **do**

  $m_{\text{ref}} \leftarrow 15 + k\,\delta$

  Take the m_{ref} lowest-SumRel sites; record $\bar{R}^2_{\text{ref}}$ and p_{ref} with random-subset bands (Algorithms 2–3)

return $\{\bar{R}^2_{ref}(m_{ref})\}$ *and* $\{p_{ref}(m_{ref})\}$

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Appendix: Theoretical Details

Between-group test. Let X_m stack the cross-product matrices (A, B, C) from an m_{ref}-site subset, and θ their population-level (N-site) counterpart. For m_{ref}, $N - m_{\text{ref}}$ both reasonably large:

$$\sqrt{m_{\text{ref}}}\,(X_{m_{\text{ref}}} - \theta) \xrightarrow{d} \mathcal{N}(0, \Sigma).$$

Since $\bar{R}^2 = g(X_{m_{\text{ref}}})$ is a smooth function, the Delta method gives

$$\sqrt{m_{\text{ref}}}\,\left(\bar{R}^2 - g(\theta)\right) \xrightarrow{d} \mathcal{N}\Big(0,\ \nabla g(\theta)^\top \Sigma\, \nabla g(\theta)\Big).$$

The empirical null from B draws, $\hat{F}_{m_{\text{ref}},B}$, approximates the true null $F_{m_{\text{ref}}}$ with error bound (DKW):

$$P\left(\sup_t |\hat{F}_{m_{\text{ref}},B}(t) - F_{m_{\text{ref}}}(t)| > \varepsilon\right) \leq 2e^{-2B\varepsilon^2}.$$

Within-group test. Fixing one m_{ref}-site set, only the cross-term $B^\pi = \mathbf{S}_\eta^\top \mathbf{T}_\eta^\pi$ varies under permutation π; A, C stay exact constants. By Hoeffding's (1951) combinatorial CLT, with exact closed-form $\mu_\pi = \mathbb{E}[T_\pi]$, $\sigma_\pi^2 = \mathrm{Var}(T_\pi)$:

$$\frac{T_\pi - \mu_\pi}{\sigma_\pi} \xrightarrow{d} \mathcal{N}(0,1) \qquad (m_{\text{ref}} \to \infty).$$

The empirical permutation null from B' draws, $\hat{G}_{m_{\text{ref}},B'}$, approximates the exact (fully enumerable at $m_{\text{ref}}!$) permutation-null G_η at the same DKW rate, $O(B'^{-1/2})$.

References: Hájek (finite-population CLT); Hoeffding (1951, combinatorial CLT); Dvoretzky–Kiefer–Wolfowitz (1956, uniform convergence rate).